CS-844 Assignment 2

Importance Sampling & Variational Inference

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Part 1: Importance Sampling

Q1. What type of problem does importance sampling help solve? What kind of integral or expectation are we trying to compute? Why might it be difficult to compute directly?

Importance sampling helps solve the problem of computing an **expectation** that is too hard to calculate directly. Specifically, we want to find the probability of observed data x by summing over all possible hidden values z:

$$p(x) = \int p(x,\, z)\, dz$$

This is **intractable** because:

- With 30 binary latent variables, the sum has $2^{30} \approx 1$ **billion** terms.
- For continuous z, the integral has **no closed-form solution**.
- Simply guessing random values of z does not work — most give near-zero probability and contribute nothing useful.

Q2. What is the core idea of importance sampling? What is the "proposal distribution"? Why do we sample from a different distribution than the one we care about?

The core idea is: instead of sampling from the hard distribution $p(z)$, we sample from a simpler **proposal distribution** $q(z)$ and then correct for the difference mathematically. The corrected estimate is:

$$p(x) \;\approx\; \frac{1}{k}\sum_{j=1}^{k} \frac{p(x,\, z_j)}{q(z_j)}, \qquad z_j \sim q(z)$$

- $q(z)$ is our **helper distribution** — easy and cheap to sample from.
- We sample from q because p is **too complex** to sample from directly.

- A good $q(z)$ should focus on values of z that are likely under $p(z \mid x)$.

Q3. What role do importance weights play? What do they correct for? How do they allow us to approximate expectations?

Each sample z_j gets an **importance weight**:

$$w_j \; = \; \frac{p(x,\, z_j)}{q(z_j)}$$

These weights **correct the mismatch** between q and p:

- If q **over-sampled** a point $\Rightarrow$ weight is **small**.
- If q **under-sampled** a point $\Rightarrow$ weight is **large**.
- The weighted average gives a **correct, unbiased estimate** of $p(x)$.

Q4. When can importance sampling perform poorly? What happens if the proposal distribution is poorly chosen?

Importance sampling fails when $q(z)$ is a **bad match** for $p(z \mid x)$:

- It misses the samples that actually matter.
- A **few samples** get extremely large weights.
- The estimate depends on 1–2 points $\Rightarrow$ **very high variance**.

Key Rule: The closer $q(z)$ is to $p(z \mid x)$, the better the result.

Part 2: Variational Inference

Q1. What problem is variational inference trying to solve? What is a posterior distribution? Why is it often intractable?

Variational inference solves the problem of computing the **posterior distribution** $p(z \mid x)$. By Bayes' rule:

$$p(z \mid x) \;=\; \frac{p(x \mid z)\, p(z)}{p(x)}$$

It is **intractable** because:

- $p(x) = \displaystyle\int p(x, z)\, dz$ is impossible for complex models.
- With 30 binary features $\Rightarrow$ 2^{30} terms to sum.
- For continuous z $\Rightarrow$ the integral has no solution.

Q2. What is the main idea behind variational inference? What is the variational family? What does it mean to "approximate" a posterior?

The main idea: pick a **simple distribution** $q_\phi(z)$ and tune it to look as close to $p(z \mid x)$ as possible.

- **Variational family** = set of simple distributions, e.g., $q_\phi(z) = \mathcal{N}(\mu, \sigma^2)$
- **"Approximate"** means: adjust $\phi = (\mu, \sigma)$ so q_ϕ resembles $p(z \mid x)$ — without computing it exactly.

Q3. How does optimization enter the picture? What is being minimized or maximized? Why is KL divergence important?

We **maximize** the **ELBO (Evidence Lower Bound)**:

$$\log p(x) \;=\; \underbrace{\text{ELBO}}_{\uparrow\,\text{maximize}} \;+\; \underbrace{D_{KL}(q_\phi \,\|\, p(z \mid x))}_{\downarrow\,\text{minimize}}$$

$$\text{ELBO} \;=\; \mathbb{E}_{q_\phi}[\log p(x, z)] + H(q_\phi)$$

- $D_{KL} = 0 \;\Rightarrow\; q_\phi$ is a **perfect match**.
- $D_{KL} > 0 \;\Rightarrow\;$ approximation has error.
- Since $\log p(x)$ is fixed, **maximizing ELBO** $\equiv$ **minimizing** D_{KL}.
- Optimization uses standard **gradient descent**.

End of Assignment 2